COMPUTER ENGINEERING

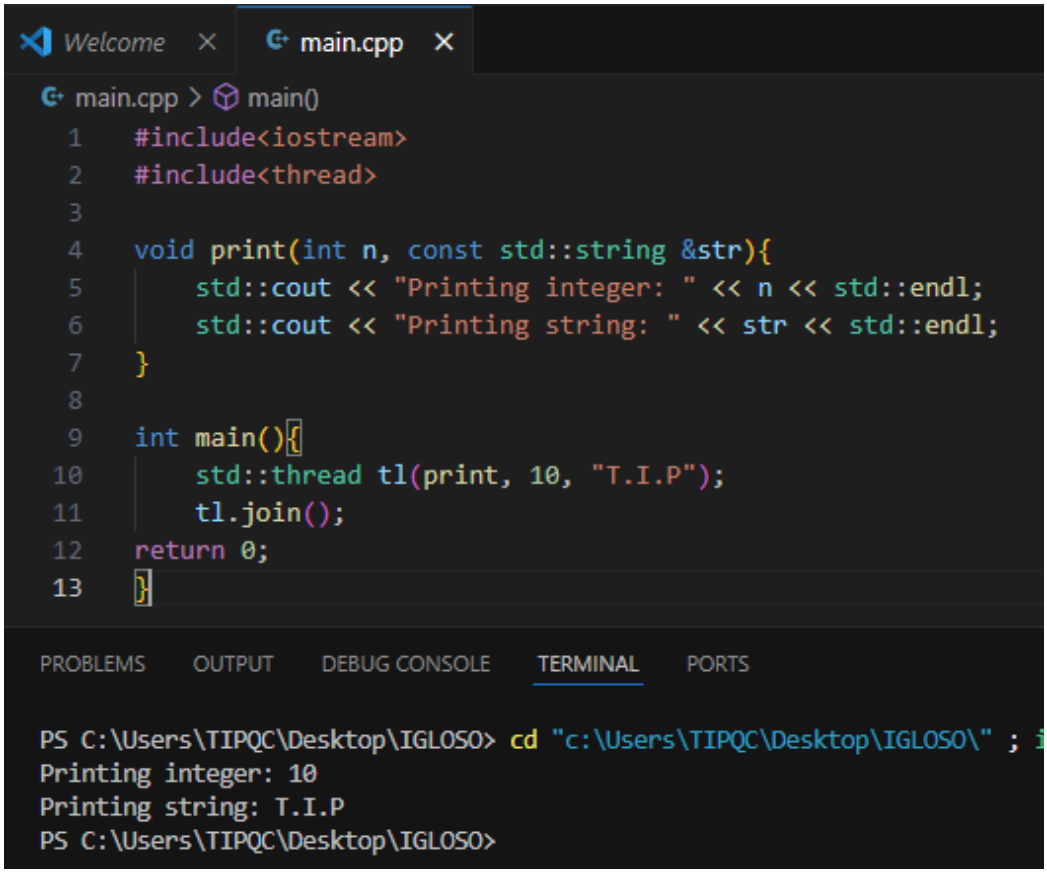
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	Oct 15, 2026	**DATE SUBMITTED:**	Oct 15, 2026
NAME:	Mike Justin Cabangon Denzen Igloso	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 13.1 PARALLELISM AND MULTITHREADING

1. PROCEDURE OUTPUT

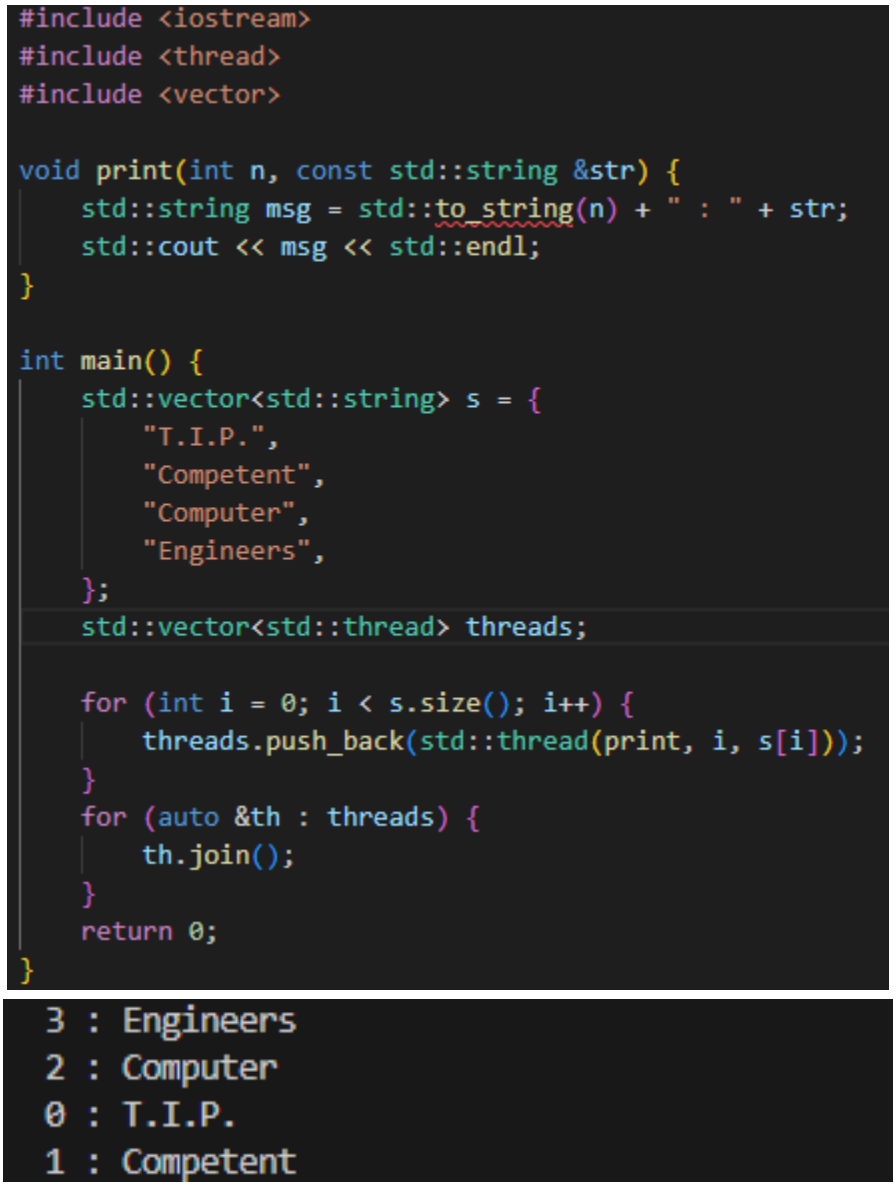
Screenshot



Analysis

For this code, the void function has a parameter of integer and string and prints it separately. So when the driver function executes std::thread called the callable functions along with the needed parameters “10, T.I.P” then prints its output.

Table 13-1. Simple One-Threaded Example

Screenshot	
Analysis	The code creates separate threads to runs each vector at the same time but the tasks finish at their own pace, they don't start and end at the same time. The code divides the tasks across multiple threads.

```cpp
#include <iostream>
#include <thread>
#include <vector>

void print(int n, const std::string &str) {
    std::string msg = std::to_string(n) + " : " + str;
    std::cout << msg << std::endl;
}

int main() {
    std::vector<std::string> s = {
        "T.I.P.",
        "Competent",
        "Computer",
        "Engineers",
    };
    std::vector<std::thread> threads;

    for (int i = 0; i < s.size(); i++) {
        threads.push_back(std::thread(print, i, s[i]));
    }
    for (auto &th : threads) {
        th.join();
    }
    return 0;
}
```

```
3 : Engineers
2 : Computer
0 : T.I.P.
1 : Competent
```

Table 13-2. Multithreaded Example.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

PART A

1. Write a definition of multithreading and its advantages/disadvantages.

NAME	ANSWER
CABANGON	
IGLOSO	Multithreading is a single program that runs multiple threads at the same time. For me, multithreading is both concurrency and parallelism because for example, the kitchen at Jollibee is like multithreading and each crew has a different kind of task. Threads are the grillman, backup, pantry, fry and stockman. The grill is cooking patty, bacon, eggs, pancakes as well as assembling different kinds of orders such as yum burger or champ jr. This is an example of concurrency where two tasks overlap. So basically each crew is a thread (parallelism) that is also doing another thread (concurrency).

2. Rationalize the use of multithreading by providing at least 3 use-cases.

NAME	ANSWER
CABANGON	
IGLOSO	

3. Differentiate between parallelism and concurrency.

NAME	ANSWER
CABANGON	
IGLOSO	

PART B

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

COMPUTER ENGINEERING

QUEZON CITY

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.